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Hemoadsorption combined with hemodialysis versus hemodialysis alone on mortality in end-stage kidney disease: a randomized, open-label, multicenter trial

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Hemoadsorption Combined with Hemodialysis versus Hemodialysis Alone on Mortality in End-Stage Kidney Disease: A Randomized, Open-label, Multicenter Trial

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Abstract

The objective of this completed, randomized, open-label trial across 11 hemodialysis centers in Shanghai, China, was to evaluate whether hemoadsorption combined with hemodialysis (HAHD) reduces mortality compared to hemodialysis (HD) alone in end-stage kidney disease patients (maintenance HD ⩾3 months, Kt/V⩾1.2). We randomized 1362 patients 1:1 to receive HAHD (n = 683) or HD alone (n = 679; mainly low-flux HD plus intermittent HDF). All 1362 randomized patients were analyzed. The primary outcome was all-cause mortality, while secondary outcomes included cardiovascular mortality and major cardiovascular events. Over a median follow-up of 39.5 months, all-cause mortality occurred in 117 (17.1%) of HAHD patients compared to 144 (21.2%) of HD patients (hazard ratio [HR]: 0.778, 95% confidence interval [CI]: 0.609–0.994; P = 0.045). HAHD also significantly reduced cardiovascular mortality (HR: 0.659, 95% CI: 0.481–0.901; P=0.009) and major cardiovascular events (HR: 0.772, 95% CI: 0.621–0.959; P=0.019). Important adverse events, primarily infections and abnormal blood pressure, were comparable between the two groups. Adding hemoadsorption significantly reduced all-cause mortality, cardiovascular mortality, and major cardiovascular events compared to HD alone (mainly low-flux HD plus intermittent HDF). Trial Registration: ClinicalTrials.gov NCT03227770.

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Introduction

End-stage kidney disease (ESKD) is a prevalent and life-threatening condition that imposes a substantial burden on patients' quality of life and healthcare systems due to its association with high mortality, morbidity and healthcare expenditures. According to the Chinese National Renal Data System (CNRDS)[1], 916,647 patients were receiving maintenance hemodialysis (HD) in China by the end of 2023. However, only 52.6%, 33.4% and 10.2% of these patients had been on dialysis for more than 3, 5 and 10 years, respectively. Despite considerable improvements in survival over the past six decades[2], high rates of complications, hospitalizations, and mortality highlight the limitations of current treatment strategies with unsatisfactory long-term outcomes[3]. These challenges are partly attributed to the retention and accumulation of uremic toxins, which appear to contribute to inflammation, anemia, and a range of osteoarticular and cardiovascular disorders.

Globally and in China, conventional hemodialysis modalities — such as low-flux hemodialysis (LFHD), high-flux hemodialysis (HFHD), and hemodiafiltration (HDF) — remain the primary blood purification techniques for ESKD patients[4]. However, these therapies rely primarily on diffusion and convection, limiting their ability to effectively remove middle-to-large molecular weight and protein-bound uremic toxins[5,6]. Despite significant technological advances, these limitations are associated with persistent morbidity, hospitalizations, diminished quality of life, and poor survival outcomes[7,8]. Furthermore, technical challenges, including those related to vascular access, can constrain the adoption of advanced methods like HDF, particularly in regions with limited healthcare resources[9].

Given the constraints of membrane-based toxin removal, there has been a growing interest in adjunctive blood purification technologies. Sorbent-based hemoadsorption represents a promising approach to enhanced toxin removal, offering a relatively safe and straightforward method to target large or protein-bound solutes that conventional dialysis fails to adequately clear[10]. Recent studies have suggested that hemoadsorption combined with hemodialysis (HAHD) could yield clinical benefits, alleviation of

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severe uremic symptoms such as chronic kidney disease-associated pruritus[11], sleep disturbances[12], microinflammation[13], dialysis-related amyloidosis[14], thereby improving patients' quality of life[15-17]. Preliminary evidence from a systematic review and meta-analysis of seven randomized controlled trials (RCTs) revealed HAHD significantly reduced overall mortality rates at 1 and 2 years compared to conventional hemodialysis[18]. However, these findings were derived from small-scale and single-center trials, limiting their generalizability.

To address this gap, we conducted a randomized, open-label, multicenter study to investigate whether HAHD could reduce mortality compared to HD alone in patients with ESKD.

Results

Patients

Between December 1, 2017 and December 31, 2019, 1419 patients were screened, and 1362 eligible patients (median age 64 years [interquartile range 54–68]; 62.2% males) were randomly assigned to receive either HAHD (n = 683) or HD (n = 679) (Figure 1). Thirteen patients (1.90%) in the HAHD group and 7 patients (1.03%) in the HD group were lost to follow-up.

Baseline characteristics, including demographic features and laboratory values were well balanced between the groups at baseline (Table 1 and Supplementary Table 1). With the exception of 4 participants in the HAHD and 5 participants in the HD groups, all participants received low-flux HD. Additionally, participants received intermittent HDF with a mean frequency of 0.77 (± 0.41) sessions per week. In the HAHD group, 614 patients (90.0%) received HAHD once every two weeks, 27 patients (3.95%) received HAHD a median of 1.4 times every two weeks (equivalent to three times a month), and 42 patients (6.15%) received HAHD twice every two weeks.

All-cause mortality

By December 31, 2021, the median duration of follow-up of the overall cohort was 39.5 months (interquartile range: 27.0–45.0). The all-cause mortality rate was 17.1% (n = 117) in the HAHD group

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compared to 21.2% (n = 144) in the HD group (hazard ratio [HR]: 0.778, 95% confidence interval [CI]: 0.609–0.994; P = 0.045; Figure 2A). Of the 261 total deaths, 163 (62.5%) were attributed to cardiovascular disease, and 69 (26.4%) to infections (Supplementary Table 2). When kidney transplantation was considered as a competing risk (HAHD: 14, HD: 5), the P value improved to 0.039 (log-rank test). Notably, the decreased mortality effect of HAHD was most pronounced in patients with a dialysis vintage of ≥ 3 years (HR: 0.612, 95% CI: 0.449–0.855) or β_2-microglobulin (β_2-MG) levels > 30mg/L (HR: 0.681; 95% CI: 0.486–0.954), whereas no significant mortality effect was observed in those with a dialysis vintage of < 3 years or β_2-MG levels ≤ 30 mg/L (Figure 2B).

Cardiovascular mortality and major cardiovascular events risk

Cardiovascular mortality was 9.66% (n = 66) in the HAHD group and 14.3% (n = 97) in the HD group (HR: 0.659, 95% CI: 0.481–0.901; P = 0.009; Figure 3). After adjusting for the competing risk of kidney transplantation, the P value improved to 0.007 (log-rank test; Figure 3A). HAHD demonstrated a greater effect on cardiovascular mortality in the subgroup of patients with a dialysis vintage of ≥ 3 years (HR: 0.394, 95% CI: 0.249–0.623) or β_2-MG levels of > 30mg/L (HR: 0.575, 95% CI: 0.372–0.887) (Figure 3B).

Additionally, there was a significant reduction in major cardiovascular events (MACE) with HAHD (HR: 0.772, 95% CI: 0.621–0.959; P = 0.019), with a P value of 0.013 (log-rank test) when considering the competing risk from kidney transplantation (Figure 4A). Subgroup analysis revealed that HAHD had a more substantial effect on reducing MACE in patients with a dialysis vintage ≥ 3 years (HR: 0.703, 95% CI: 0.525–0.941) and/or β_2-MG levels > 30mg/L (HR: 0.645, 95% CI: 0.481–0.865), (Figure 4B).

Sensitivity analyses

The sensitivity analyses among Per Protocol (PP) population (660 in the HAHD group and 667 in the HD group) showed a consistent reduction in risk for CVD-specific mortality and MACE, but a marginal

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reduction in all-cause mortality from HAHD (Table 2). We further post-hoc adjusted for sex and found a consistent reduction risks for mortality (all-cause and CVD mortality) and MACE.

Laboratory tests

Changes in baseline plasma levels of laboratory measurements during 24-month follow-up are presented in Supplementary Table 3. HAHD resulted in a lower accumulation of β_2-MG and homocysteine compared to HD. Additionally, median changes from baseline were more pronounced in the HAHD group for high-sensitivity C-reactive protein, ferritin, creatine kinase, creatine kinase-MB, and pro B-type natriuretic peptide.

Safety

Adverse events were reported in 47 (6.88%) of 683 patients in the HAHD group and 64 (9.42%) of 679 patients in the HD group (Table 3). Of these, 3 events (two episodes bleeding, one of abnormal blood pressure) in the HAHD and 2 events (two infections) in the HD group were reported as serious adverse reactions believed to be related to treatment. Changes in baseline plasma levels of safety indices over the 24-month follow-up indicated that median changes in baseline levels of white blood cell, red blood cell, platelets, aspartate aminotransferase (AST), alanine aminotransferase (ALT), albumin, and glucose showed no differences between the two groups (Supplementary Table 3).

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Discussion

In this randomized, open-label, multicenter study involving 1362 ESKD patients with a median of 39.5 months follow-up demonstrated that conventional HD (mainly low-flux HD plus intermittent HDF) plus hemoadsorption at least once every two weeks significantly reduced the risks of all-cause and cardiovascular-specific mortality, and MACE compared to conventional HD alone. These findings were consistent across various subgroups of patients, with the greatest benefits observed in patients with

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longer dialysis vintage and elevated β_2-MG levels, underscoring HAHD's potential for managing high-risk populations.

To date, our study is one of the largest randomized trials to assess the effects of HAHD versus HD on mortality. Our findings are consistent with the 1-year and 2-year survival benefits reported in a recent meta-analysis that pooled data from seven small-sample, single-center RCTs conducted in China[18]. Prospective observational studies further support these results. Notably, a large multicenter cohort study in China, involving 4,623 patients, demonstrated that HAHD significantly reduced all-cause mortality compared to HD alone[19]. Together, these findings strengthen the evidence that HAHD may offer meaningful improvements in overall survival for patients with ESKD.

The reduction in all-cause mortality observed in our study was most notably linked to the significant decrease in cardiovascular-related mortality and the risk of MACE. Cardiovascular diseases are a leading cause of death in patients undergoing dialysis, and our study demonstrated that HAHD offers a powerful approach to mitigating this risk. Several mechanisms may explain this benefit. First, HAHD has been shown to improve blood pressure control, left ventricular mass index, epoetin dosage, and hemoglobin levels[17,20]. It also reduces microinflammatory markers and advance glycation end products, all of which contribute to a reduction in the incidence of cardiovascular events, such as atherosclerosis, heart failure, arrhythmia, and angina pectoris[16,21,22]. These results are in line with other studies, which have demonstrated that HAHD improves cardiovascular health through multiple mechanisms, including the enhanced removal of uremic toxins and improved microinflammation[13,18]. Our findings further strengthen the evidence that HAHD can improve cardiovascular outcomes in this high-risk population.

Importantly, our study observed that the reduction in mortality was particularly pronounced in patients with longer dialysis vintage and elevated β_2-MG levels. This suggests that HAHD may be especially beneficial high-risk patients who have accumulated greater levels of uremic toxins over time. These results are supported by prior research showing that HAHD provides protective effects in patients with

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advanced stages of kidney disease, such as those with elevated β_2-MG, a marker of uremic toxin accumulation[11,23]. Our findings highlight that HAHD could be an effective strategy for managing patients with more severe disease, thereby improving their survival and quality of life.

The study findings have significant implications for clinical practice, particularly in regions where HFHD and HDF are inaccessible due to technical, economic, or resource limitations. While HFHD is considered the standard of care in Western countries, its benefits for mortality remain inconclusive[24,25], and its widespread adoption is hindered by high costs and stringent requirements. The CNRDS reports that 61.9% of dialyzers used in China are of the low-flux variety, and the average blood flow rate is only 235 mL/min[26], limiting the feasibility of HDF, which requires a convective volume over 23 L per session and a blood flow rate exceeding 350 ml/min[27,28]. In this context, HAHD represents an accessible and cost-effective alternative. Our cost-effectiveness analysis[29] revealed that HAHD incurs an incremental cost-effectiveness ratio of RMB 174,486 (USD 25,251) per quality-adjusted life-year (QALY), well within the willingness-to-pay threshold of three times the Gross Domestic Product (RMB 212,676 or USD 30,778). This positions HAHD as a viable option for improving survival outcomes in resource-constrained environments.

Despite the promising results, this study has limitations. First, it is important to clarify that the HDF regimens used in the conventional HD group reflect common clinical practice in China, which may involve a lower frequency and intensity than the high-volume, thrice-weekly HDF that is standard in many Western countries and recent major trials. This difference in the comparator treatment limits the direct generalizability of our findings to settings where high-efficiency HDF is the routine standard of care. The observed all-cause mortality risk in our cohort was lower than typically reported, potentially due to selection bias favoring patients with better prognoses. Additionally, residual renal function and certain clinical symptoms, such as vomiting or syndrome control, were not systematically monitored. Furthermore, we did not collect data on hemoadsorption use prior to enrollment; while this is a

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limitation, a mandatory 1-month washout period was implemented before randomization to mitigate potential legacy effects from prior therapies. The hemoadsorption duration in our study was limited to two hours per session, consistent with existing Chinese guidelines, but emerging evidence from other countries[30,31] suggests that extending hemoadsorption to four hours may yield additional benefits. Further studies are needed to explore these possibilities. Blinding of participants and clinicians to treatment allocation was not performed due to the inherent technical differences between the HAHD and HD procedures. This study lacked an independent adjudication committee, and event verification was performed by local investigators. Although predefined criteria and standardized training were used, and efforts were made to maintain blinding, the absence of centralized, blinded adjudication may introduce potential bias in event classification and adverse event reporting. A significant baseline imbalance in sex distribution was observed. Although sex-adjusted analyses showed consistent results, residual confounding cannot be entirely excluded. Also, the observed reduction in effect size in the PP population compared to modified intention-to-treat (mITT) population may be due to reduced sample size, loss of statistical power, and potential selection bias from excluding non-adherent participants. In particular, patients who discontinued treatment (e.g., due to intolerance or perceived improvement) may differ systematically from those who remained fully adherent. These factors may attenuate the apparent treatment effect, although the consistency in direction across analyses supports the robustness of our findings. Additionally, while the study period (2019–2021) overlapped with the COVID-19 pandemic, policies for life-sustaining therapies ensured that primary endpoint data remained complete. However, the pandemic severely disrupted the collection of subjective data, leading to substantial and non-random loss of Quality of Life (QoL) information. An analysis of this incomplete data was therefore deemed scientifically unsound, and its omission is acknowledged as a significant limitation. Lastly, the achieved sample size was slightly below the target due to recruitment challenges during the COVID-19 pandemic. However, the lower-than-expected dropout rate maintained sufficient statistical power.

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In conclusion, our findings indicate that HAHD significantly reduces all-cause mortality, cardiovascular-specific mortality, and MACE risks compared to conventional HD alone. These results highlight the potential of HAHD to improve survival outcomes in ESKD patients, particularly in high-risk populations. The slightly smaller effect size in the per-protocol population may reflect selection bias from excluding nonadherent patients. Future research should further explore the mechanisms underlying these benefits and evaluate extended hemoadsorption durations to optimize therapeutic outcomes.

Methods

Study design

This randomized, open-label, multicenter study evaluated the effects of HAHD versus HD alone on all-cause mortality, cardiovascular mortality and MACE in patients with ESKD. The study protocol, previously published in detail[32], was carried out across 11 hemodialysis centers in Shanghai, China. The study adhered to Good Clinical Practice guidelines and the principles of the Declaration of Helsinki. The study protocol was approved by the Ethics Committees at each of the 11 participating study sites: the Ethics Committee of Xinhua Hospital affiliated to Shanghai Jiaotong University School of Medicine, the Ethics Committee of Shanghai Changhai Hospital, the Ethics Committee of Longhua Hospital, Shanghai University of Traditional Chinese Medicine, the Ethics Committee of Renji Hospital affiliated to Shanghai Jiao Tong University School of Medicine, the Ethics Committee of Yangpu Hospital, School of Medicine, Tongji University, the Ethics Committee of Shanghai Sixth People's Hospital Affiliated to Shanghai JiaoTong University School of Medicine, the Ethics Committee of Tongji Hospital, School of medicine, Tongji University, the Ethics Committee of Zhongshan Hospital, Fudan University, the Ethics Committee of Shanghai General Hospital, Shanghai Jiaotong University, the Ethics Committee of Shanghai Ninth People's Hospital affiliated to Shanghai Jiao Tong University School of Medicine, and the Ethics Committee of Rui Jin Hospital affiliated to Shanghai Jiao Tong University School of Medicine. All participants provided written informed consent before enrollment.

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The study was registered with ClinicalTrials.gov (NCT03227770). The reporting of this study conforms to the CONSORT 2010 guidelines (Supplementary Note 1).

Participants

Participants were recruited from December 1, 2017, to December 31, 2019. Eligible participants were adults aged 18 ≥ years, with a diagnosis of ESKD, undergoing regular HD treatment for at least 3 months, with a standard Kt/V ≥ 1.2. Exclusion criteria included: white cell count < 4×10^9 cells/L or platelet count < 100×10^9 cells/L, cerebral hemorrhage within the previous twelve weeks, MACE within the previous eight weeks, severe heart failure (New York Heart Association class IV), active gastrointestinal bleeding or coagulation disorders, active malignancy or infection, pregnancy or breastfeeding, and significant cognitive impairment.

Randomization and interventions

Following a 1-month washout period during which all enrolled participants received LFHD, HFHD, or HDF, eligible participants were randomly assigned in a 1:1 ratio to receive HAHD or a conventional dialysis regimen (described below). Randomization was stratified by study center, and the randomization sequence was generated by an independent statistician with allocation concealed in opaque envelopes. Allocation concealment was ensured by revealing group assignments only after baseline data registration. At each clinical center, unmasked study personnel prepared treatments, but investigator and participant remained masked to the allocation prior to enrollment.

In the HAHD group, participants received conventional HD supplemented hemoadsorption at least once every two weeks. Hemoadsorption was performed using an HA130 cartridge (Jafron Biomedical, Zhuhai, China), placed in series before the filter. Cartridges were prepared following the manufacturer's instruction. Each cartridge was primed with 12,500 IU of heparin, gently shaken and rotated, and allowed to rest for at least 30 minutes. They were subsequently primed with 2,000-3000 mL of normal saline after being connected to the filter. During HAHD sessions, the HA130 cartridge was integrated

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into the HD circuit for the first 2 hours with a blood flow rate of 150-200 mL/min. The blood flow was increased to 200–250 mL/min after discontinuing the HA cartridge at the 2-hour mark, to optimize the efficacy of the subsequent HD-only phase.

In the conventional HD group, participants received LFHD, HFHD, or HDF based on their pre-washout dialysis regimen, with a minimum treatment duration of 10 h/week (two 5-hour sessions or three 4-hour sessions per week). Polysulfone filters with a surface area of 1.4–1.8 m^2 were utilized. HD prescription included the following parameters: dialysis time: 4–5 h, blood flow rate: 200-300 mL/min, average dialysate flow rate: 500 mL/min, bicarbonate dialysate with calcium ion concentrations of 1.25–1.5 mmol/L. For HDF, treatments adhered to the Standard Operating Procedures for Blood Purification (Chinese Edition)[33]: blood flow rate: > 250 mL/min, dialysate flow rate: 500-800 mL/min, pre-dilution volume: 30–50 L, or a post-dilution volume: 15-25 L. Low molecular weight heparin or unfractionated heparin served as anticoagulants, with dosages tailored to patient body weight. The baseline dialysis prescription was intended to be maintained throughout the study for all participants. Any adjustments were infrequent and not systematically different between the groups, thus minimizing the risk of bias influencing the outcomes.

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Outcomes

The primary outcome of study was all-cause mortality, while secondary outcomes included cardiovascular mortality and MACE. These outcomes were initially assessed over a minimum planned follow-up period of 24 months. Due to slow initial recruitment, the enrollment period was extended until December 2019. The observation period concluded in December 2021, ensuring that all participants had a planned follow-up of at least 24 months, with the earliest enrollees having a maximum follow-up duration of 49 months.

MACE was defined as the occurrence of any of the following events: angina pectoris, myocardial infarction, severe arrhythmia, progression of heart failure, pericarditis, myocarditis, cardiac or

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peripheral vascular surgery, stroke, or peripheral vascular disease. The mortality of the patients was documented by attending physicians at the respective hospital or reported by relevant parties and subsequently verified by physicians. Causes of death were classified in accordance with China's National Disease Surveillance Points (DSP) system, which provides standardized and reliable mortality reporting. The timing and cause of death, as well as MACE occurrences, were documented by treating physicians, and all classifications were reviewed and verified by study investigators. In addition, information on primary kidney disease and comorbidities was carefully reviewed and confirmed using medical records.

Safety monitoring was conducted throughout the study. Given that the most severe safety outcomes (death and MACE) were analyzed as the primary and secondary endpoints, the reporting of other adverse events was focused on those deemed by investigators to be potentially related to the treatment procedure, thereby providing a targeted assessment of procedural safety.

Laboratory assessments were performed at baseline and at 6, 12, 18, and 24 months after study initiation (Supplementary Table 4). The measured parameters included renal and metabolic markers (intact parathyroid hormone [iPTH], β_2-MG, blood urea nitrogen, serum sodium, potassium, calcium, phosphate, and glucose), inflammatory markers (high-sensitivity C-reactive protein and homocysteine), liver function tests (total bilirubin, direct bilirubin, alanine aminotransferase, aspartate aminotransferase, γ-glutamyltransferase, alkaline phosphatase, total protein, and albumin), cardiac biomarkers (troponin I, myoglobin, lactate dehydrogenase, hydroxybutyrate dehydrogenase, and pro B-type natriuretic peptide, creatine kinase, creatine kinase-MB), and dialysis efficiency (standard Kt/V).

Statistical analysis

We determined that the enrollment of 1364 patients (200 primary outcome events) would provide 80% power to determine a 30% relative risk reduction in mortality with HAHD compared with the HD at a

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two-sided type I error of 5% and a follow-up duration of 2 years. Further details on the power calculation have been previously reported[32].

The intention-to-treat analysis, which included all randomized patients, was used to evaluate baseline characteristics. For the analyses of primary and secondary outcomes, a mITT set was employed, consisting of all patients who received at least one treatment session. The per-protocol population, comprising all patients who adhered to the prescribed treatment plan, was used for sensitivity analyses. Follow-up was defined as the time interval from the time of randomization to the date of death, loss to follow-up, withdrawal, or December 31, 2021, whichever occurred first.

A Cox proportional hazards regression model adjusted for study center was used to estimate HRs and 95% CIs for all-cause and cardiovascular mortality and MACE. Kaplan–Meier analysis with the log-rank test was used to compare time to event for all-cause and cardiovascular mortality and MACE rate between the HAHD and HD groups. For primary and secondary outcomes, all analyses were performed with kidney transplantation defined as a competing event. Follow-up was right-censored at the time of transplantation, and deaths occurring after transplantation were not included, as they could not be reliably attributed to the effects of dialysis treatment. We graphically assessed the model assumption of proportionality using log cumulative hazard plots and by examining scaled Schoenfeld residuals. The proportional hazards assumption was met for all outcomes.

We additional performed subgroup analysis to assess for possible heterogeneity of treatment effect. Predefined subgroups were based on age (< 65 and ≥ 65 years), sex (males and females), dialysis vintage (< 3 and ≥ 3 years), baseline β_2-MG levels (≤ 30 and > 30 mg/L), cause of kidney failure (chronic glomerulonephritis, hypertension nephrosclerosis, diabetic nephropathy, polycystic kidney, and others), baseline ferritin levels (< 151 and ≥ 151 μg/L), baseline homocysteine levels (< 30.8 and ≥ 30.8 μmol/L), baseline pro B-type natriuretic peptide levels (< 2421 and ≥ 2421 pg/L), and baseline iPTH (< 600 and ≥ 600 pg/mL).

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All analyses were performed using the SAS 9.4 software (SAS Institute, Cary, NC).

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Data availability

The study protocol including statistical analysis plan is available as Supplementary Note in the Supplementary Information. The datasets generated and analyzed during the current study are not publicly available due to institutional and patient privacy regulations but are available from the corresponding author (jianggengru01@xinhuamed.com.cn) for academic research purposes one year after publication. Access is subject to approval by the Ethics Committee of Xinhua Hospital. Requests will be responded to within 20 working days. Requests for access to de-identified data will be reviewed for compliance with data sharing policies and ethical considerations, and approved data will be shared via secure transfer mechanisms. Source data containing the numerical values underlying the survival curves and forest plot are provided with this paper.

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Author contributions

W.Lu and G.Jiang conceived and designed the study. W.Lu and X.Zhang accessed and verified the data, performed statistical analyses, and drafted the initial manuscript. H.Huang, G.Ji, Z.Guo, Y.Deng, Z.Ni, R.Zhou, N.Wang, C.Yu, X.Ding, Q.Yu, Y.Wang, and X.Chen represented the coordinating centers and were responsible for patient recruitment, clinical management, and data collection. The corresponding author (W.Lu) had full access to all study data and assumed final responsibility for the decision to submit the manuscript for publication. All authors critically reviewed the manuscript and approved the final version for submission.

Competing interests

The authors declare no competing interests.

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Table 1. Baseline characteristics

Characteristics	All patients (*n* = 1362)	HAHD (*n* = 683)	HD (*n* = 679)
Age, years	61 (54, 68)	60 (54, 67)	63 (54, 70)
Males, n (%)	847 (62.2)	458 (67.1)	389 (57.3)
Body weight, kg	62.3 ± 11.7	63.6 ± 11.3	61.0 ± 12.0
BMI, kg/m^2	22.4 ± 3.59	22.6 ± 3.60	22.1 ± 3.57
SBP, mmHg	143 ± 21.0	143 ± 21.0	143 ± 21.0
DBP, mmHg	78.2 ± 12.7	78.3 ± 12.7	78.2 ± 12.7
Heart rate, beats/min	78.3 ± 9.28	78.1 ± 9.05	78.6 ± 9.50
Cause of renal failure			
Chronic glomerulonephritis	533 (39.1)	259 (37.9)	274 (40.4)
Hypertension nephrosclerosis	95 (6.98)	50 (7.32)	45 (6.63)
Diabetic nephropathy	160 (11.8)	87 (12.7)	73 (10.8)
Polycystic kidney	48 (3.52)	25 (3.66)	23 (3.39)
Others	82 (6.02)	40 (5.86)	42 (6.19)
NYHA class			
I	335 (43.0)	175 (43.4)	160 (42.4)
II	365 (46.8)	192 (47.6)	173 (45.9)
III	78 (10.0)	35 (8.68)	43 (11.4)
IV	2 (0.26)	1 (0.25)	1 (0.27)
History of chronic diseases			
Diabetes	165 (12.1)	91 (13.3)	74 (10.9)

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Dyslipidemia	960 (79.3)	486 (80.1)	474 (78.6)
Hypertension	176 (12.9)	97 (14.2)	79 (11.6)
Cardio- or cerebro-vascular diseases	183 (13.4)	102 (14.9)	81 (11.9)
Anemia	1175 (86.3)	591 (86.5)	584 (86.0)
HGB, 10^{12}/L	110 ± 16.6	111 ± 16.3	110 ± 16.8
Albumin, g/L	39.4 (37.0, 42.0)	39.5 (37.0, 41.8)	39.1 (37.0, 42.0)
Hs-CRP, mg/L	2.39 (0.84, 6.68)	2.28 (0.76, 6.64)	2.40 (0.89, 6.69)
SpKt/V	1.40 ± 0.36	1.36 ± 0.32	1.44 ± 0.38
iPTH, pmol/L	27.4 (13.8, 48.0)	27.6 (13.8, 48.3)	27.2 (13.8, 45.6)
β_2-MG, mg/L	36.4 ± 15.8	37.1 ± 15.2	35.7 ± 16.4
WBC	6.10 (5.07, 7.43)	6.19 (5.10, 7.46)	6.03 (5.06, 7.43)
RBC	3.66 ± 0.59	3.69 ± 0.56	3.62 ± 0.61
BPC	188 ± 64.3	192 ± 64.3	184 ± 64.1
Vintage of hemodialysis, year	3 (1, 7)	4 (2, 8)	3 (1, 6)
Types of dialysis			
HD	255 (18.7)	113 (16.5)	142 (20.9)
HD + HDF	1107 (81.3)	570 (83.5)	537 (79.1)
HD sessions per week	2.22 ± 0.40	2.22 ± 0.39	2.23 ± 0.41
HDF sessions per week	0.77 ± 0.41	0.78 ± 0.39	0.75 ± 0.42
Single-session dialysis time (h)	3.97 ± 0.15	4.00 ± 0	4.01 ± 0.09
Total sessions per week	2.99 ± 0.09	3.00 ± 0	2.98 ± 0.13
Total dialysis time (h/week)	11.98 ± 0.19	12.0 ± 0	11.97 ± 0.26

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Blood flow (mL/min)	228 ± 21	227 ± 21	228 ± 21
Dialysate flow (mL/min)	508 ± 27	506 ± 25	510 ± 30

Note: BMI: body mass index; SBP: systolic blood pressure; DBP: diastolic blood pressure; NYHA: New York Heart Association; HGB: hemoglobin; Hs-CRP: high sensitivity C-reactive protein; SpKt/V: single pool Kt/V; iPTH: intact parathyroid hormone; β_2-MG: β_2-microglobulin; WBC: white blood cell; RBC: red blood cell; BPC: blood platelet count; HD: hemodialysis; HDF: hemodiafiltration; HAHD: hemoadsorption combined with hemodialysis.

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Table 2. Primary and secondary outcomes

Outcomes	mITT Population (HAHD vs. HD: 683 vs. 679)			PP Population (HAHD vs. HD: 660 vs. 667)	
	Rates (HAHD / HD, %)	HR (95% CI)*	Sex-Adjusted HR (95% CI)**	Rates (HAHD / HD, %)	HR (95% CI)
Primary outcome					
All-cause mortality	17.1/21.2	0.778 (0.609, 0.994)	0.759 (0.593, 0.971)	17.7/21.1	0.810 (0.634, 1.036)
Secondary outcomes					
CVD-specific mortality	9.66/14.3	0.659 (0.481, 0.901)	0.632 (0.461, 0.866)	10.0/14.2	0.688 (0.502, 0.943)
Risk of MACE	21.4/27.3	0.772 (0.621, 0.959)	0.751 (0.604, 0.935)	21.8/27.3	0.790 (0.635, 0.984)

*Hazard Ratios (HRs) and 95% Confidence Intervals (CIs) were estimated using a Cox proportional hazards regression model adjusted for study center. ** HRs (95% CIs) were estimated further adjusted for

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sex. Kidney transplantation was defined as a competing event. CVD, Cardiovascular Disease; MACE, major adverse cardiovascular events; mITT: modified intent-to-treat; PP: per-protocol.

Table 3. Adverse events

Adverse events	**HAHD (*n* = 683)**	**HD (*n* = 679)**
Total	47	64
Infections	21	24
Respiratory tract infections	14	17
Urinary tract infections	3	5
Biliary tract infections	4	2
Bleeding	6	4
Gastrointestinal bleeding	4	2
Urinary tract bleeding	2	2
Abnormal blood pressure	7	22
Hypertension	4	14
Hypotension	3	8
Skeleton fracture	0	3
Others	13	11

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Figure legends

Figure 1. Patient enrollment and follow-up.

Figure 2. All-cause mortality. (A) Kaplan-Meier analysis of all-cause mortality. Shaded areas represent 95% confidence intervals. (B) Subgroup analysis of all-cause mortality. Data are presented as Hazard Ratios (measure of center) and 95% confidence intervals (error bars). HR: hazard ratio; CI: confidence interval.

Figure 3. Deaths related to major cardiovascular events. (A) Kaplan-Meier analysis of cardiovascular mortality. Shaded areas represent 95% confidence intervals. (B) Subgroup analysis of cardiovascular mortality. Data are presented as Hazard Ratios (measure of center) and 95% confidence intervals (error bars). HR: hazard ratio; CI: confidence interval; CVD: cardiovascular disease.

Figure 4. Major cardiovascular events. (A) Kaplan-Meier analysis of major cardiovascular events. Shaded areas represent 95% confidence intervals. (B) Subgroup analysis of cardiovascular events. Data are presented as Hazard Ratios (measure of center) and 95% confidence intervals (error bars). HR: hazard ratio; CI: confidence interval; CVD: cardiovascular disease.

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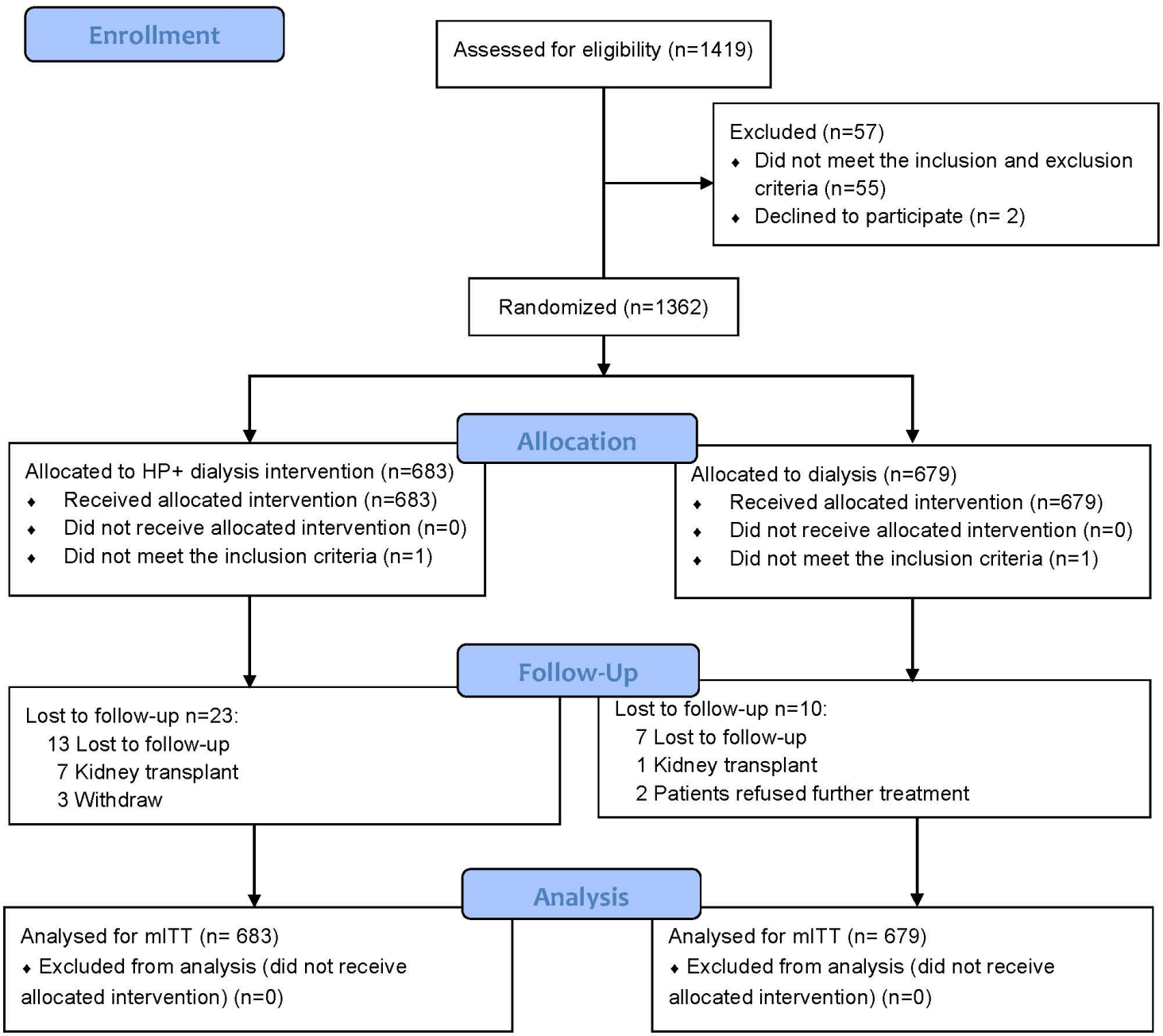
Editor's Summary:

This randomized open-label trial shows that hemoadsorption combined with hemodialysis reduces all-cause and cardiovascular mortality, and cardiovascular events in patients with end-stage kidney disease compared to hemodialysis (mainly low-flux hemodialysis plus intermittent hemodiafiltration) alone.

Peer Review Information: *Nature Communications* thanks Gonzalo Ramírez-Guerrero, Brendan Smyth, Tom Greene, Haojin Zhou and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.


Enrollment
Assessed for eligibility (n=1419)
Excluded (n=57)
Did not meet the inclusion and exclusion criteria (n=55)
Declined to participate (n= 2)
Randomized (n=1362)
Allocation
Allocated to HP+ dialysis intervention (n=683)
Received allocated intervention (n=683)
Did not receive allocated intervention (n=0)
Did not meet the inclusion criteria (n=1)
Allocated to dialysis (n=679)
Received allocated intervention (n=679)
Did not receive allocated intervention (n=0)
Did not meet the inclusion criteria (n=1)
Follow-Up
Lost to follow-up n=23:
13 Lost to follow-up
7 Kidney transplant
3 Withdraw
Lost to follow-up n=10:
7 Lost to follow-up
1 Kidney transplant
2 Patients refused further treatment
Analysis
Analysed for mITT (n= 683)
Excluded from analysis (did not receive allocated intervention) (n=0)
Analysed for mITT (n= 679)
Excluded from analysis (did not receive allocated intervention) (n=0)

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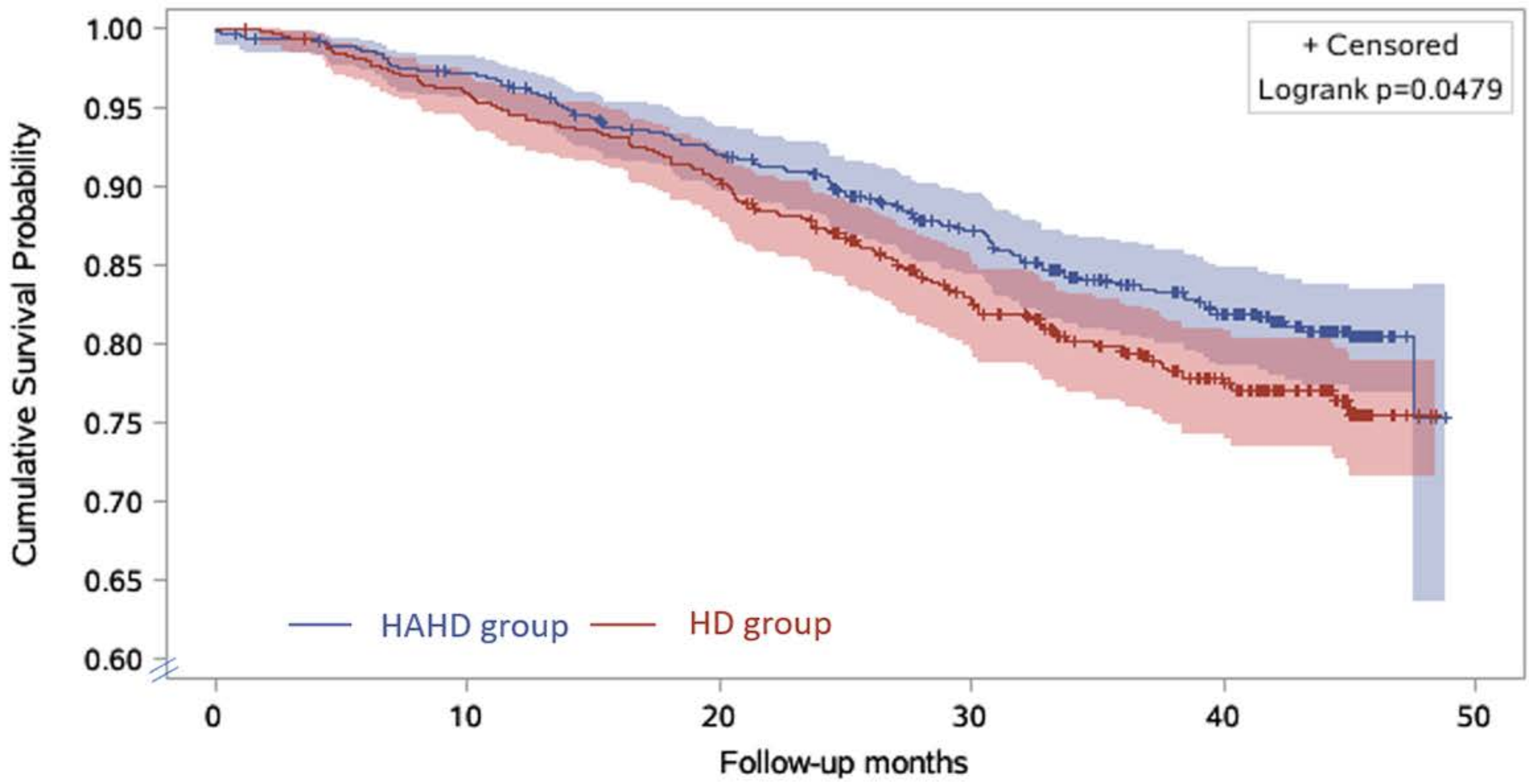

HAHD group	683	659	615	493	358	0
HD group	679	649	612	459	314	0

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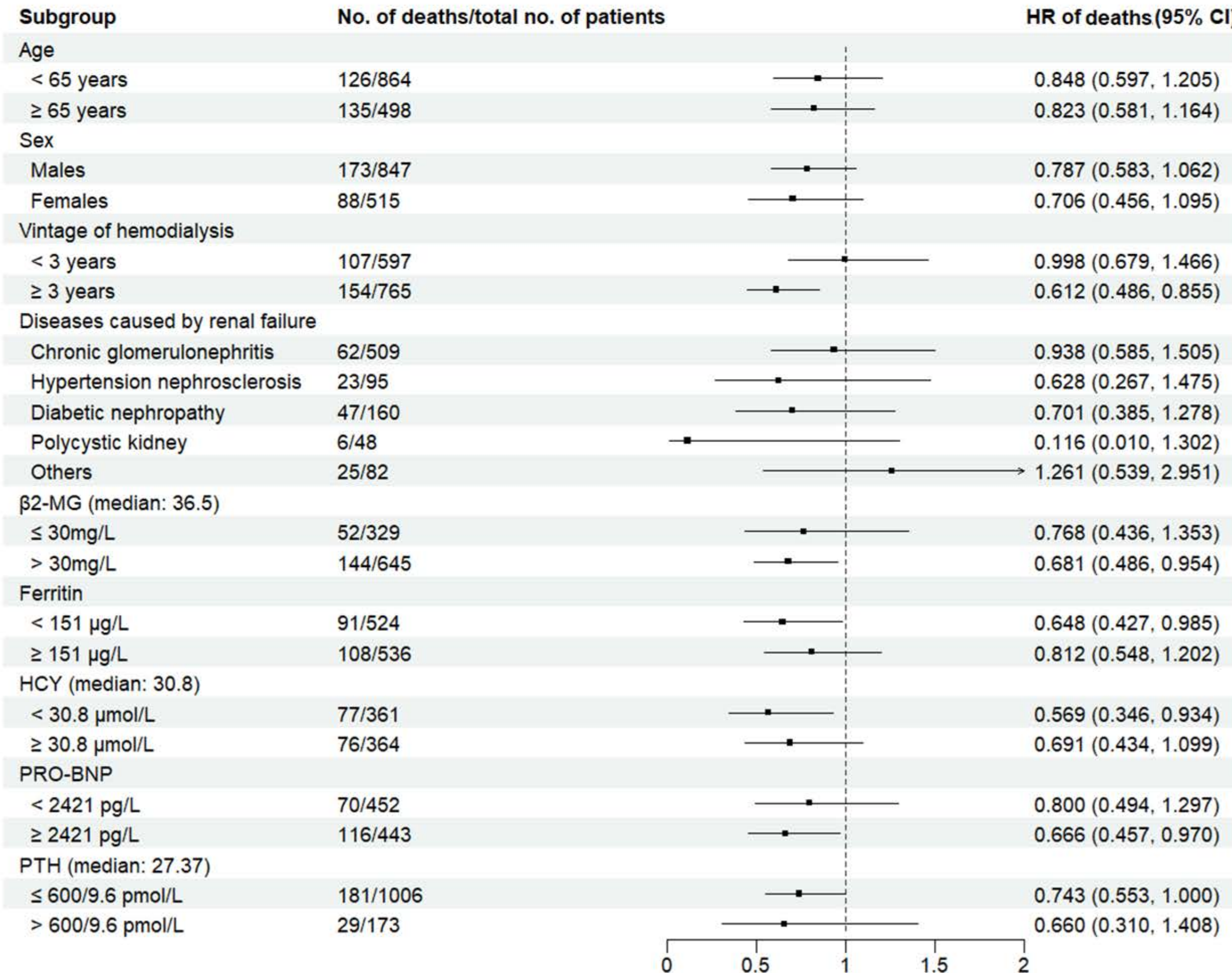

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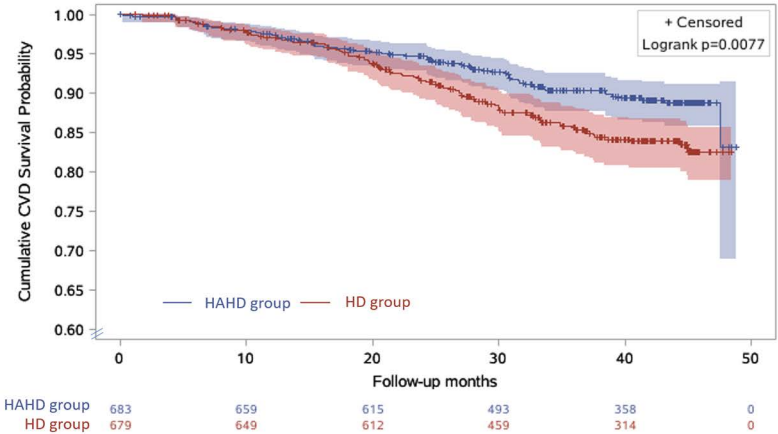
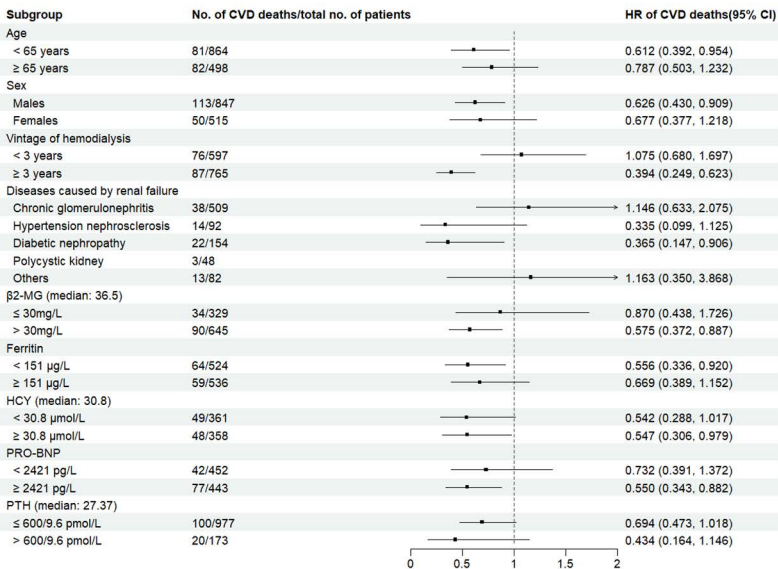

+ Censored
Logrank p=0.0077
Cumulative CVD Survival Probability
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
HAHD group
HD group
0
10
20
30
40
50
Follow-up months
HAHD group 683 659 615 493 358 0
HD group 679 649 612 459 314 0

B


Subgroup
No. of CVD deaths/total no. of patients
HR of CVD deaths(95% CI)
Age
< 65 years 81/864 0.612 (0.392, 0.954)
≥ 65 years 82/498 0.787 (0.503, 1.232)
Sex
Males 113/847 0.626 (0.430, 0.909)
Females 50/515 0.677 (0.377, 1.218)
Vintage of hemodialysis
< 3 years 76/597 1.075 (0.680, 1.697)
≥ 3 years 87/765 0.394 (0.249, 0.623)
Diseases caused by renal failure
Chronic glomerulonephritis 38/509 1.146 (0.633, 2.075)
Hypertension nephrosclerosis 14/92 0.335 (0.099, 1.125)
Diabetic nephropathy 22/154 0.365 (0.147, 0.906)
Polycystic kidney 3/48
Others 13/82 1.163 (0.350, 3.868)
β2-MG (median: 36.5)
≤ 30mg/L 34/329 0.870 (0.438, 1.726)
> 30mg/L 90/645 0.575 (0.372, 0.887)
Ferritin
< 151 μg/L 64/524 0.556 (0.336, 0.920)
≥ 151 μg/L 59/536 0.669 (0.389, 1.152)
HCY (median: 30.8)
< 30.8 μmol/L 49/361 0.542 (0.288, 1.017)
≥ 30.8 μmol/L 48/358 0.547 (0.306, 0.979)
PRO-BNP
< 2421 pg/L 42/452 0.732 (0.391, 1.372)
≥ 2421 pg/L 77/443 0.550 (0.343, 0.882)
PTH (median: 27.37)
≤ 600/9.6 pmol/L 100/977 0.694 (0.473, 1.018)
> 600/9.6 pmol/L 20/173 0.434 (0.164, 1.146)
0
0.5
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1.5
2

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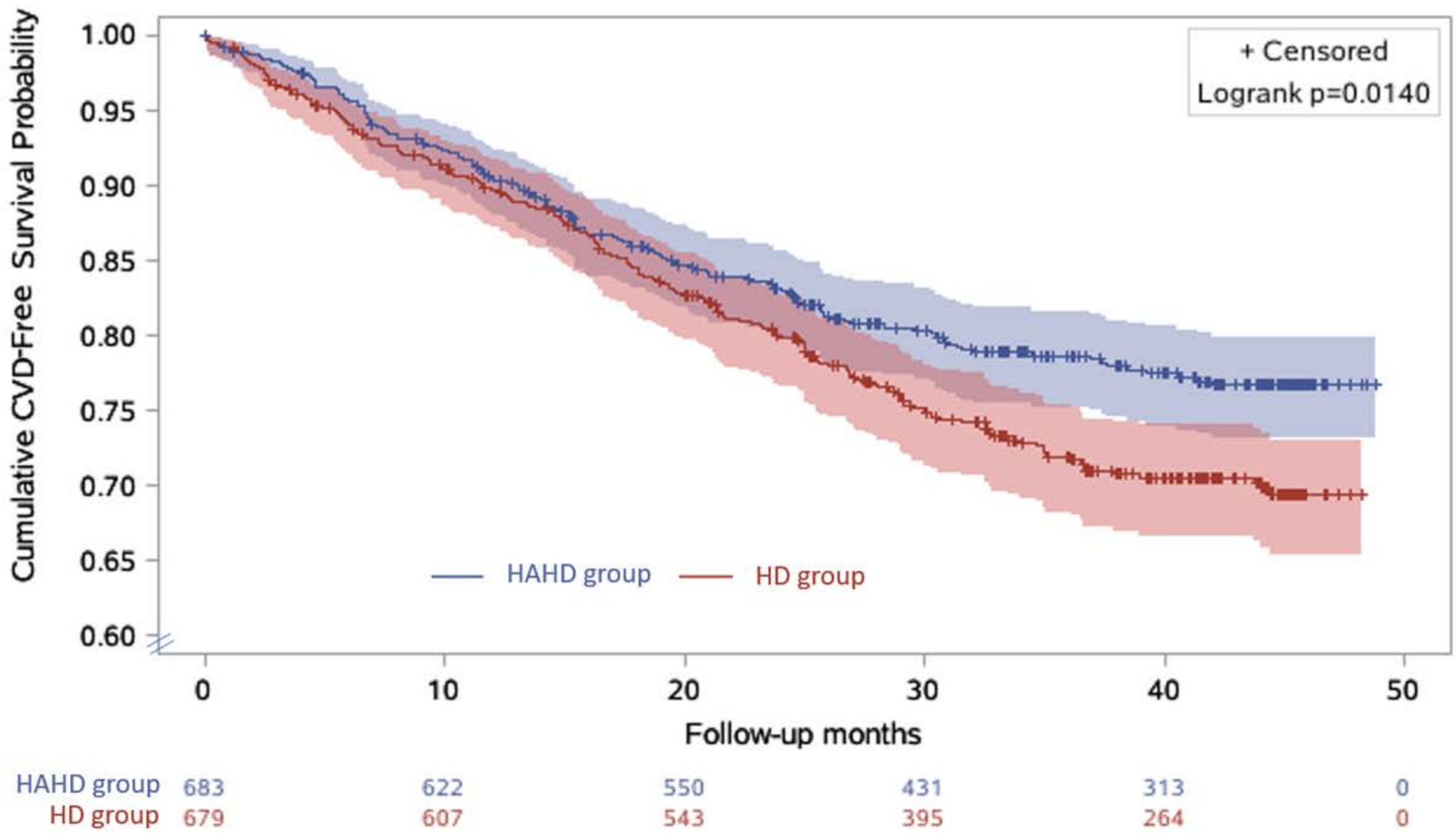
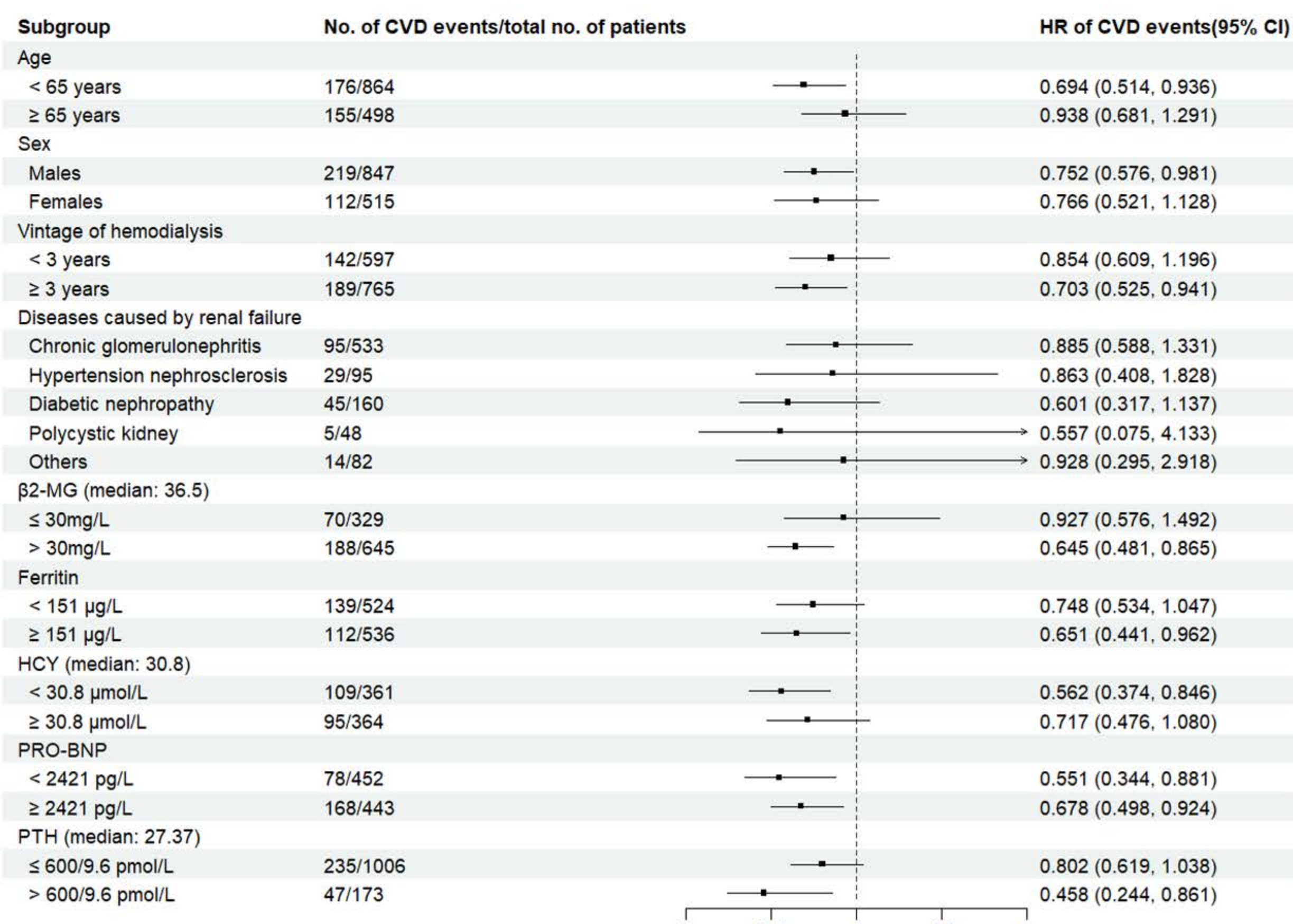

Cumulative CVD-Free Survival Probability
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
+ Censored
Logrank p=0.0140
HAHD group
HD group
0
10
20
30
40
50
Follow-up months
HAHD group 683 622 550 431 313 0
HD group 679 607 543 395 264 0

B


Subgroup
No. of CVD events/total no. of patients
HR of CVD events(95% CI)
Age
< 65 years 176/864 0.694 (0.514, 0.936)
≥ 65 years 155/498 0.938 (0.681, 1.291)
Sex
Males 219/847 0.752 (0.576, 0.981)
Females 112/515 0.766 (0.521, 1.128)
Vintage of hemodialysis
< 3 years 142/597 0.854 (0.609, 1.196)
≥ 3 years 189/765 0.703 (0.525, 0.941)
Diseases caused by renal failure
Chronic glomerulonephritis 95/533 0.885 (0.588, 1.331)
Hypertension nephrosclerosis 29/95 0.863 (0.408, 1.828)
Diabetic nephropathy 45/160 0.601 (0.317, 1.137)
Polycystic kidney 5/48 0.557 (0.075, 4.133)
Others 14/82 0.928 (0.295, 2.918)
β2-MG (median: 36.5)
≤ 30mg/L 70/329 0.927 (0.576, 1.492)
> 30mg/L 188/645 0.645 (0.481, 0.865)
Ferritin
< 151 μg/L 139/524 0.748 (0.534, 1.047)
≥ 151 μg/L 112/536 0.651 (0.441, 0.962)
HCY (median: 30.8)
< 30.8 μmol/L 109/361 0.562 (0.374, 0.846)
≥ 30.8 μmol/L 95/364 0.717 (0.476, 1.080)
PRO-BNP
< 2421 pg/L 78/452 0.551 (0.344, 0.881)
≥ 2421 pg/L 168/443 0.678 (0.498, 0.924)
PTH (median: 27.37)
≤ 600/9.6 pmol/L 235/1006 0.802 (0.619, 1.038)
> 600/9.6 pmol/L 47/173 0.458 (0.244, 0.861)
0
0.5
1
1.5
2